Standard form

1 Express the following equations in standard form.

a $y = 2x - 5$ **b** $y = \frac{2x}{5} + 4$ **c** $y = 6x - 5$ **d** $y = \frac{6x}{5} - 6$

e $y = \frac{-x}{-6}$ **f** $y - 5x = 5$

2 Luke wanted to rewrite $y = \frac{9x}{10} + 4$ in standard form. His steps are as follows:

1. $y = \frac{9x + 40}{10}$
2. $10y = 9x + 40$
3. $9x + 10y = 40$

State which step Luke made a mistake.

3 For each of the following equations of lines:

a Solve for x-value of the x-intercept of the line.

b Solve for y-value of the y-intercept of the line.

c Graph the equation of the line.

a $5x - 3y = -15$ **b** $4x - 3y = -12$ **c** $2x + y = 10$ **d** $2x - 3y = 6$

e $5x + y = 1$

4 For each of the following equations of lines:

i Express the equation in slope-intercept form.

ii Determine the slope of the line.

iii State the y-value of the y-intercept of the line.

a $4x - 12y = 4$ **b** $6x - y = -8$ **c** $2x + 5y = 35$

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5 Consider the slope of a line that passes through the given point:

i By substituting into the equation $y = mx + b$, find the value of b for this line.

ii Write the equation of the line in standard form.

a slope $= \frac{7}{3}$, point $(7, -9)$

b slope $= 2$, point $\left(5, \frac{-9}{5}\right)$.

6 A straight line passes through the point $\left(0, \frac{1}{5}\right)$ with slope 4.

a Find the equation of the line in the form $y = mx + b$.

b Express this equation in standard form.

c Solve for the x-value of the x-intercept.

7 Consider the lines with the following equations:

- Line A: $4x + 3y = -5$ and
- Line B: $7x + 9y = 1$

a Express Line A in slope intercept form:

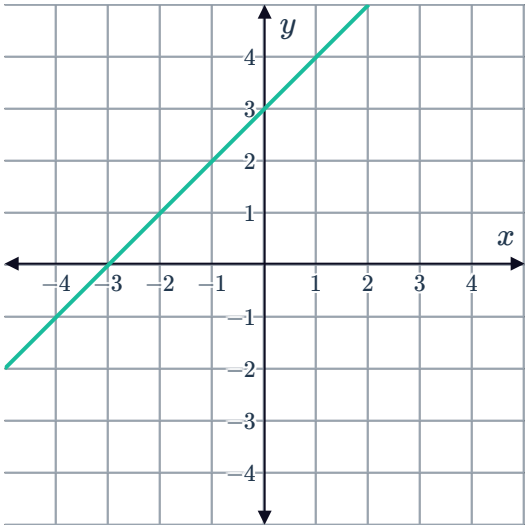
b Express Line B in slope intercept form.

c Determine the slopes of the lines A and B.

d Identify the steeper line.

8 Which line is steeper, $5x + 5y = -1$ or $5x + 2y = 5$?

9 Write the equation of the graphed line.



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10 Determine whether the y-intercept of the straight line is represented by the constant in the following equations.

a $-5y = 5x - 1$ **b** $y = -5 + x$ **c** $-5x - 5y = -5$ **d** $y = 2x - 2$

11 The equation of a straight line is given by $x - 5y = -2$. Solve for the y-value of the y-intercept.

12 The point $(-2, -3)$ lies on the line $ax + 3y = 1$.

a Find the value of a.

b Determine the slope of the line.

13 Consider the following statement:
Five times the $x-$ value, added to three times the $y-$ value is -4.

a Write the statement as an equation in two variables.

b Graph the equation.

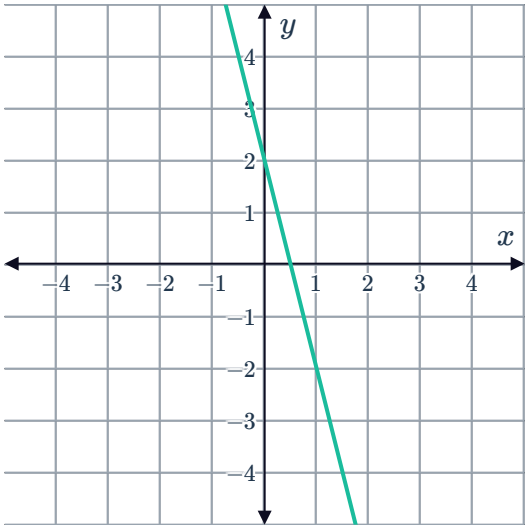
14 Consider the graph of the line:

a Determine the y-intercept

b Determine the slope of the line.

c Find the equation of the line in the form $y = mx + b$.

d Express the equation of the line in standard form.



15 The line $kx + 3y = 15$ passes through the point $(13, -8)$.

a Find the value of k.

b Solve for the x-value of the x-intercept of the line.

c Solve for the y-value of the y-intercept of the line.

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Additional questions

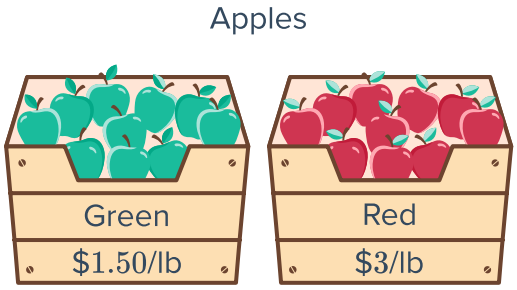
16 Salvina needs to buy a mix of red and green apples. She is going to spend $\$15$ in total. Green apples cost $\$1.50$/lb and red apples cost $\$3$/lb.

a Write a linear equation in standard form to represent the scenario. Use the variables x for green and y for red.

b Calculate and interpret the x-intercept of the linear equation from part (a).

c Determine the number of green apples she could buy if she bought 2 red apples.

d State the domain of the linear function.



17 Write an expression for the slope, m, of the line represented by the equation $Ax + By = C$. State any restrictions. Explain how this expression might be helpful.

18 Describe a scenario where using the standard form of a linear equation is more useful than the slope-intercept form.

19 Compare and contrast the standard form and slope-intercept form of a linear equation. Include how they are similar, different and the benefits of each form.

20 Effie is a entomologist and is currently studying mosquitos and spiders. She knows that mosquitos have six legs and spiders have eight legs.
In her lab, she has a mix of mosquitos and spiders. Between all the bugs, there is a total of 240 legs.

a Write an equation to represent this scenario.

b Draw a graph of the linear equation.

c State and describe the domain of the equation.

d Explain whether or not every point on the line represents a possible solution.

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21 Nakisha is selling the baby outfits her daughter has outgrown and is buying second-hand toddler outfits. She is able to sell each baby outfit for $\$3$ and she buys each toddler outfit for $\$5$. In the end, she wants to make a profit of exactly $\$30$.
Profit $=$ revenue $-$ cost
Revenue is the money earned
Cost is the money spent

a Write an equation to represent the scenario, using x for the number of baby outfits sold, y for the number of toddler outfits bought, and the profit of exactly $\$30$.

b Calculate and interpret the x-intercept.

c Nakisha has 25 baby outfits available to sell and needs 15 toddler outfits. Determine if she is able to make her desired profit of $\$30$.

d Using the additional information from part (c), state and interpret the domain restriction.

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